
The 75% Envelope Test

How Pitkin County's thermal envelope demolition calculation works, what counts toward it, and how to draw it so a plans examiner can verify it.

JURISDICTION

Pitkin County, CO
Building Department

CODE BASIS

2021 IECC R503.1.1
as amended by county

GOVERNING POLICY

Existing Building Thermal
Envelope Policy, 1/28/2026

WORKED EXAMPLE

Thermal Envelope Example
Plan (Sheets A.101-A.103)

WHY THE CALCULATION EXISTS

One threshold decides whether your remodel is treated as a new house

The demo calculation sheets are not a demolition plan in the usual sense. They are an area accounting exercise whose only job is to prove which side of 75% your project falls on.

2021 IECC § R503.1.1, EXCEPTION 3

Alterations exposing or demolishing more than 75 percent of the building's thermal envelope area shall comply with the energy requirements of Section R407 (new dwelling requirements).

Under 75%, ordinary alteration rules apply: envelope assemblies you touch comply with R402, and exposed envelope is brought to prescriptive code minimums unless the Chief Building Official grants a hardship. Over 75%, the *entire* dwelling has to meet new-construction energy requirements under R407, with REMP obligations following.

LOCAL WORDING DIFFERS FROM THE BASE CODE

Pitkin County rewrote R503.1.1 wholesale. The county’s adopted Exception 3 reads “alterations **exposing** more than 75% of the building’s thermal envelope area” — it drops “or demolishing.” The CBO’s policy memo quotes the base IECC text with both verbs. In practice the local test is exposure-centric, which matters for scopes that disturb assemblies without removing them.

The county amendment also requires pre- and post-construction Energy Rating Index assessments (ANSI/RESNET/ICC 301) on all alterations, to confirm the home does not perform worse than before. Like-for-like replacement work is excepted from the ERI requirement, provided building energy use is not increased.

SHEET STRUCTURE

Three surfaces, three sheets, one combined percentage

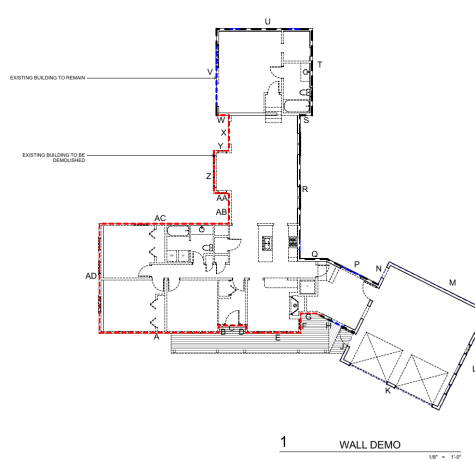
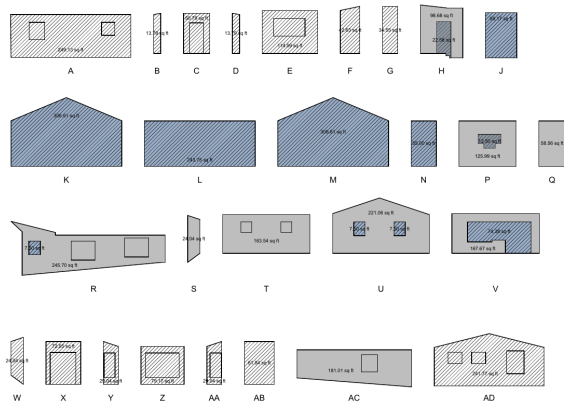
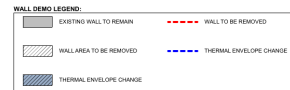
The county’s example splits the thermal envelope into its three surface types — one sheet each — and repeats the same running total on every sheet so the reviewer can check the math from any page.

SHEET	SURFACE	TOTAL SF	REMOVED SF	%
A.101	Roof — flat plane	3,248.01	2,444.38	75.3%
A.102	Wall	3,324.01	2,169.72	65.3%
A.103	Floor	2,636.39	1,712.44	65.0%
—	Combined envelope	9,208.41	6,326.54	68.70%
Sample Residence, 123 Pitkin St., Aspen. Every figure above reproduces from the sheet tables exactly.				

The combined number governs. Roof alone is 75.3% — over the line on its own — but the test sums all three surfaces, so the project passes at 68.70%. Do not compute the percentage surface by surface and stop at the first one that fails.

DEMO CALCULATION NOTES:

1. DEMOLITION CALCULATIONS BASED ON THERMAL ENERGY PER TON COUNTY BUILDING CODE SECTION 1611.1. DISCREPANCY 1. ALSO DISCREPANCY ON DISCREPANCY MORE THAN 75% OF THE BUILDING THERMAL ENVELOPE AREA SHALL COMPLY WITH THE ENGINEER REQUIREMENTS OF SECTION 1601.
2. PER LAND-USE CODE CHAPTER 111 - DEFINITIONS, REBUILDING MEANS THE RENOVATION OF AN EXISTING STRUCTURE THAT DOES NOT EXCEED THE MAXIMUM GROUND-LEVEL ELEVATION OF THE FOOTPRINT OF THE STRUCTURE. BY THE USE OF THE STRUCTURE, THE FLOOR AREA OF THE STRUCTURE IS MAINTAINED.
3. PER LAND-USE CODE CHAPTER 111 - DEFINITIONS, REPLACEMENT MEANS THE COMPLETE REMOVAL OF ALL OR A PORTION OF A STRUCTURE AND SUBSTITUTION OF THE ORIGINAL STRUCTURE WITH A NEW STRUCTURE. A REPLACEMENT STRUCTURE MAY CHANGE THE ORIGINAL SIZE AND LOCATION OF THE FOOTPRINT OF THE STRUCTURE.



SAMPLE RESIDENCE

123 PITKIN ST.
ASPEN, CO 81611

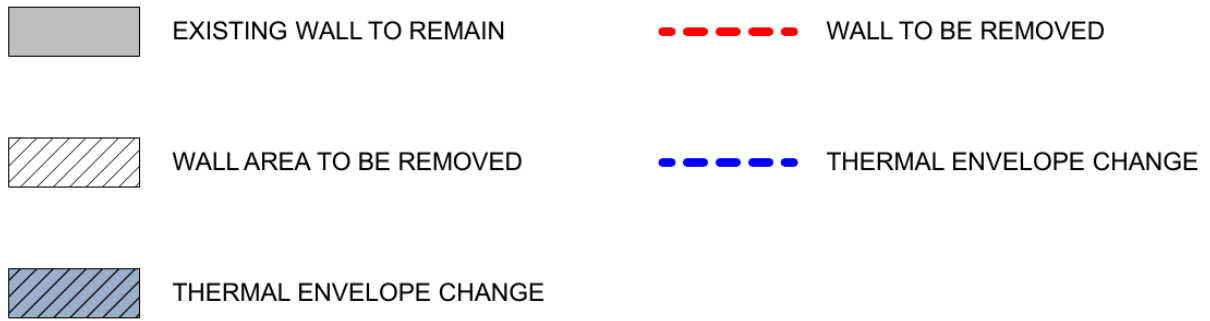
PROJECT NO:	2601
ISSUE:	DATE:
PERMIT	1/04/2026

DEMO CALCULATIONS

A.102

- 01 **Unfold the exterior walls into flat elevations** and cut them into discrete panels, one per plane. The example uses 28 panels lettered A through AD — every jog and every gable face is its own panel.
- 02 **Tag every panel with its square footage** on the drawing, and carry the same letters into a table with two columns: *Individual Wall Area* and *Area of Wall to be Removed*.
- 03 **Color-code each panel** against the legend, and key the same conditions onto the floor plan with dashed linework.
- 04 **Count partial areas as partials.** Panel H is 98.68 sf with only 22.58 sf counted; R is 245.70 sf with 7.50 sf; U is 221.06 sf with 15.00 sf. You tally the disturbed portion, not the whole panel.
- 05 **Sum the panels, then roll the wall totals into the combined figure** with roof and floor, and print that combined total on all three sheets.

WALL DEMO LEGEND:



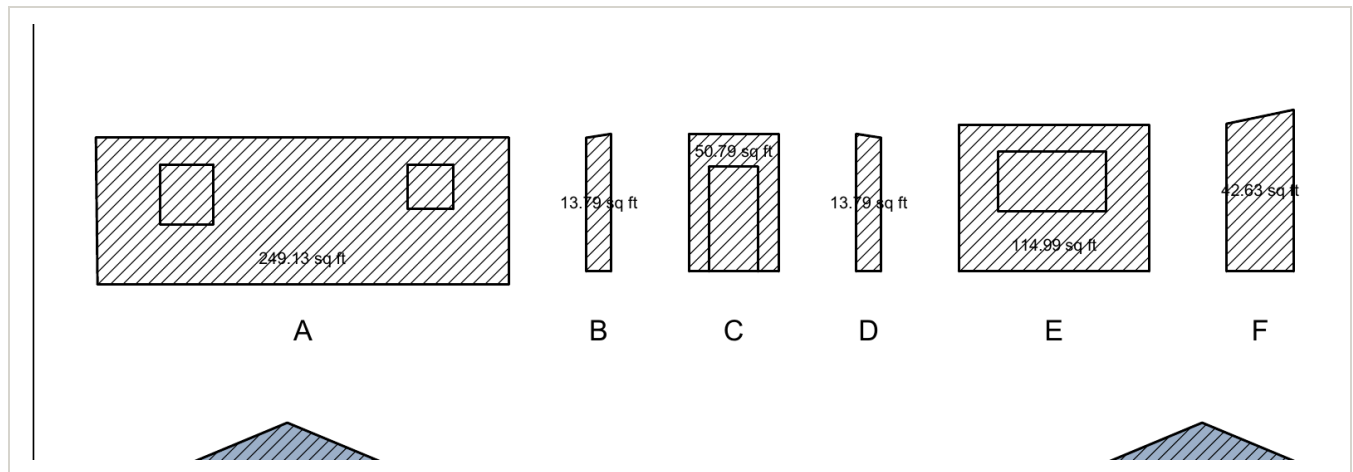
THE LEGEND DOES REAL WORK

Three fills apply to the unfolded elevation panels; two dashed line types apply to the plan. Note that “wall to be removed” and “wall *area* to be removed” are deliberately separate — the second is an assembly that is disturbed while the wall itself stays.

COUNTING RULE

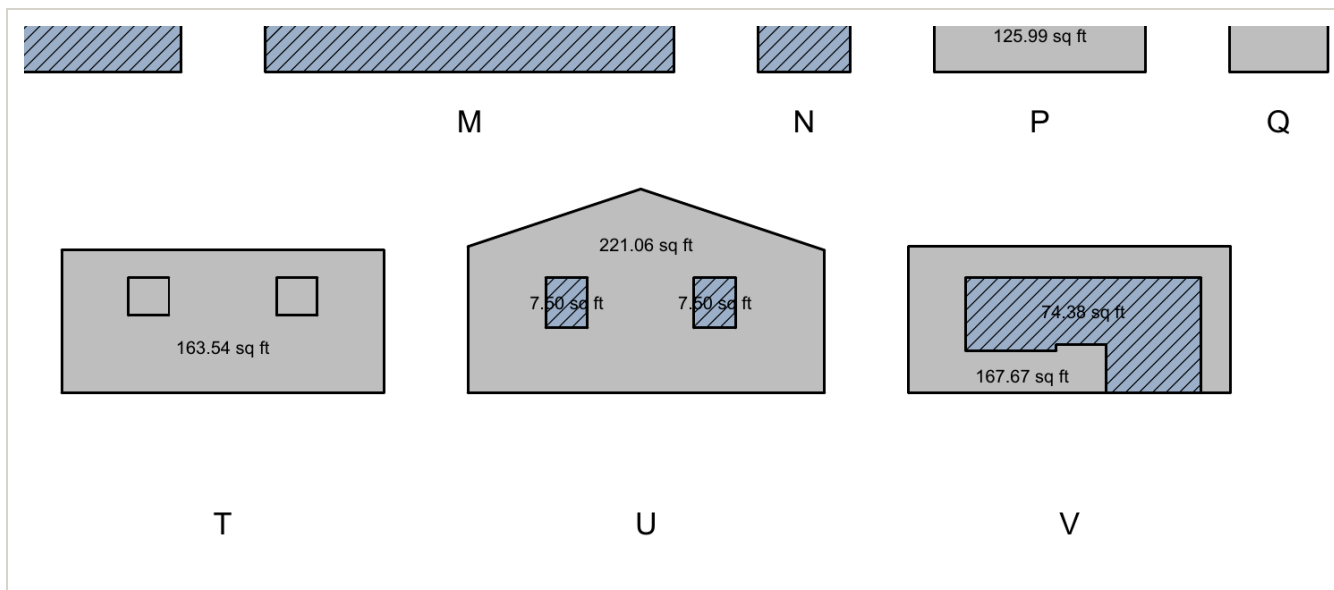
Walls with windows are counted gross

Openings are **not** deducted. Both columns — total area and removed area — are gross flat-plane elevation areas with glazing included. The county’s own drafting settles it three different ways.



PANELS A–F — HATCH RUNS THROUGH THE OPENINGS

Panel A is one number, 249.13 sf, with two windows drawn inside it. The hatch is continuous behind the openings and the outlines sit on top. Same on E (114.99 sf) and C (50.79 sf).



PANELS T, U, V – WHEN A WINDOW GETS ITS OWN NUMBER

T (163.54 sf) has two untouched windows and no figures on them. U is a 221.06 sf gross gable that stays, with two changing windows hatched and tagged 7.50 sf each — and the table shows 15.00 sf removed for panel U. V carries a 74.38 sf partial area inside a 167.67 sf panel.

QUANTITY	BASIS
Individual Wall Area the denominator	Gross flat-plane elevation area. Window, door and garage door openings are not deducted. Gable ends measured as the true elevation shape, triangle included.
Area to be Removed the numerator	Same gross basis. If the wall goes, its full gross area counts, glazing included.
A window on its own	Gets its own square footage only when the window is the element changing in a wall that otherwise stays. It already sits inside its panel's gross area, so it is added to the removed column only — never to the total.

Why gross is the defensible basis

Per Chapter 2 of the 2021 IECC, the building thermal envelope is the set of assemblies that *enclose conditioned space*. A window is an envelope assembly with its own U-factor in Table R402.1.3 — not a hole in the envelope. Gross is also the only basis a reviewer can verify by scaling your elevation. Above all it must be consistent: the result is a ratio, so the same basis has to feed both columns.

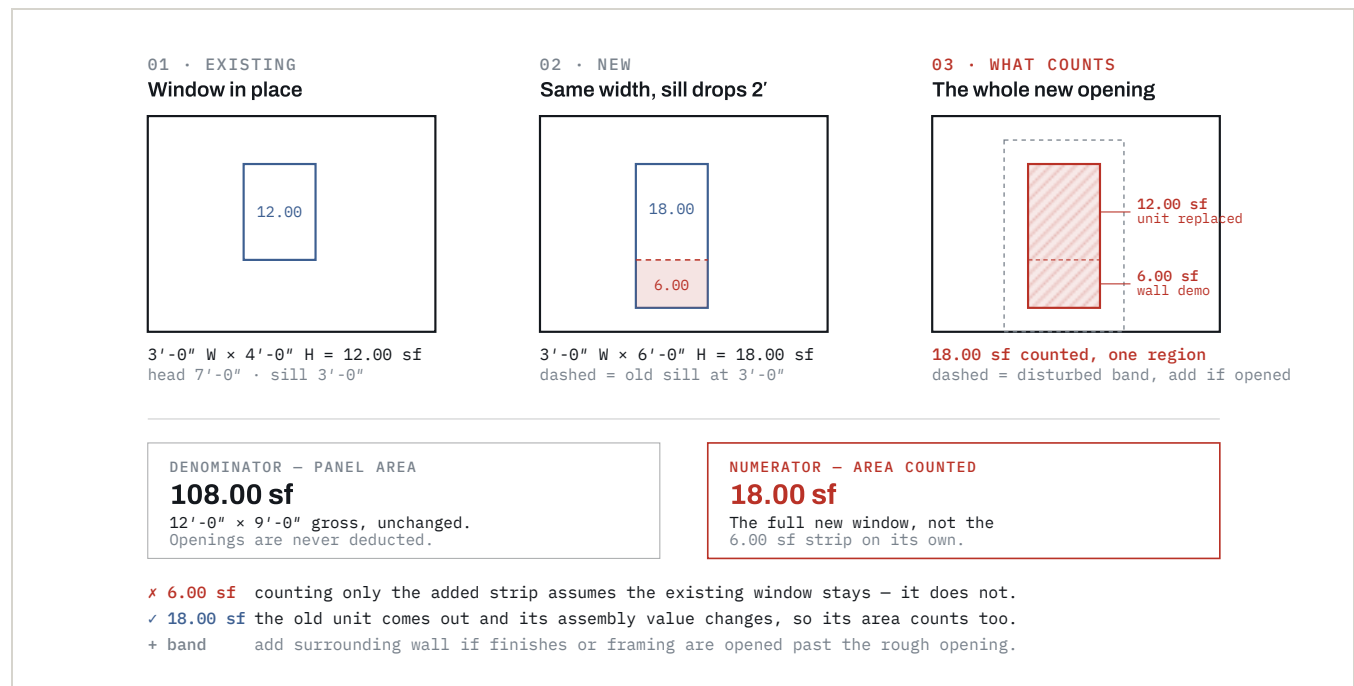
MEASUREMENT CONVENTIONS TO HOLD

- Measure to the outside of siding — the county's example states this convention on the same sheet.
- Flat plane, per plane. No unfolding shortcuts that merge two orientations into one panel.
- Do not double count. The 3,324.01 sf wall total is the sum of gross panel areas and nothing else.

WORKED CASE

Enlarging an opening: the whole new window counts

Replacing an existing window with a taller unit of the same width — does the existing window's area count, or only the strip of wall being cut out below the old sill? The full area of the new opening counts.



3'-0" x 4'-0" REPLACED WITH 3'-0" x 6'-0"

Count 18.00 sf, not the 6.00 sf strip. Dimensions illustrative.

Why the old window's area comes with it

- The existing unit is removed. Panel U on the county's example counts two changed windows at their full areas — 7.50 sf each — in a wall that otherwise stays. The precedent counts whole windows, never a difference between old and new.

- **The added strip is wall demolition on its own terms.** Cutting the sill down means removing sheathing and framing and setting a new header or cripples: cavity exposed, area counted.
- **They are one region on the drawing.** Hatch the new opening as a single area and tag it once, the way panel V carries a 74.38 sf region inside a 167.67 sf panel — don't split it into two line items.

THE DENOMINATOR NEVER MOVES

Because openings are not deducted, enlarging a window changes nothing about the panel's gross area. The wall was 108.00 sf before and 108.00 sf after. Only the numerator moves — which is the whole reason this question matters.

Add the disturbed band when you open past the opening

A new header usually means opening interior finishes and framing beyond the rough opening. That opened area is exposed envelope under the policy's interior-finish rule, so hatch the region you actually open, not just the window rectangle, and tag that larger area.

SCOPE AT AN OPENING	COUNT
Taller or wider unit in an enlarged opening	Full area of the new opening, plus any wall opened around it
Same-size unit swapped, wall untouched	Full window area, as a thermal envelope change — unless the permit is limited to window, door, siding or veneer replacement only
New opening cut in solid wall	Full area of the new opening, plus the opened band
Opening infilled with wall	Full area of the old opening — the assembly and its values change

Separately from the percentage, the new unit has to comply with R402 fenestration requirements as an altered assembly, and if every window and all the siding on that side are replaced, R503.1.1.1 pulls that whole side into Table R402.1.3.

CONFIRM THIS ONE

The county's example plan does not include an enlarged-opening case, so the treatment above applies its two established precedents — whole-window counting on panel U, region counting on panels V, H and R. Counting the full opening is both the defensible reading and the conservative one, and at these areas the cost of being conservative is small next to the risk of a change order pushing a project past 75% after approval.

What counts as disturbance — including things you never touch

Exposing counts as demolishing

From the CBO’s policy: *“Removal of interior wall finishes that expose wall and ceiling cavities even if existing framing remains shall be considered disturbance of the building thermal envelope and shall be included in the thermal envelope depiction and calculations.”* Gut the drywall and that wall is in your numerator, framing intact or not. This is why the legend separates “wall to be removed” from “wall area to be removed.”

Unheated to heated conversion counts

Panels J, K, L, M and N carry an asterisk on the example sheet: *“Area counted as demolition due to conversion from non-heated space to heated space & thermal envelope values change.”* Bringing a garage, crawlspace or porch inside the envelope — or changing an assembly’s R-value — puts that area in the count even where no framing is disturbed.

981.14 SF COUNTED FROM CONVERSION ALONE	45% OF THE ENTIRE WALL DEMO COUNT	68.70% PROJECT TOTAL — ONLY 6.3 PTS OF HEADROOM
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Like-for-like exterior siding does not count

Replacing exterior wood siding with the same product, without opening the cavity and without changing assembly values, stays out of the numerator. Three provisions agree.

- **The policy carves it out.** Depiction plans are required for all envelope alterations “except for permits limited to exterior veneer, siding, window, or door replacement only,” and such replacement “does not, by itself, trigger new dwelling requirements or require exposure of interior wall cavities.”
- **The disturbance rule is about interior finishes.** Pulling siding gets you to sheathing, not into the cavity.
- **The county’s amended R503.1.1 Exception 1** covers “the removal and replacement or the covering of existing materials, elements, equipment or fixtures using new materials, elements, equipment or fixtures that serve the same purpose.”

Three ways it starts counting

TRIGGER	STATUS	WHY
Sheathing comes off too	COUNTS	The cavity is exposed. That area lands in the numerator regardless of what happens to the framing.
All siding <i>and</i> all windows on a side	COUNTS	R503.1.1.1 applies: that side must comply with Table R402.1.3, sized against the demonstrated existing cavity R-value. Assembly values change, so the area is treated as a thermal envelope change.
Not actually like-for-like	COUNTS	Added continuous insulation, or any rainscreen or detail that shifts the assembly U-factor or R-value, is a thermal envelope value change.
Siding only, cavity closed, values unchanged	EXCLUDED	Exception 1 like-for-like replacement. Also exempt from the ERI assessment requirement, provided energy use is not increased.

ONE AMBIGUITY WORTH COVERING

The policy’s construction-document section asks for the area “proposed to be **altered**, exposed, or demolished.” “Altered” is broader than the code’s “exposing,” and an examiner could read a siding swap as an alteration. When siding work sits inside a larger project that already needs a depiction plan, draw those walls in the “existing wall to remain” fill, keep their square footage in the denominator, and carry a note:

SIDING REPLACEMENT IS LIKE-FOR-LIKE. NO CAVITY EXPOSURE AND NO CHANGE TO ASSEMBLY U-FACTOR OR R-VALUE. NOT COUNTED AS EXPOSED THERMAL ENVELOPE AREA PER R503.1.1 EXCEPTION 1.

The memo is recent — dated 1/28/2026 — so this wording has little review history. Confirm the reading with the plans examiner before submitting.

DON'T CONFUSE THESE

The linear-foot numbers are a different test

The same wall sheet carries linear-foot tallies that have nothing to do with the 75% energy threshold:

TALLY	REMAIN	DEMOLISHED	%
Linear exterior wall	209 LF	139 LF	39.9%
Second tally, labeled “wall area” but given in LF	125 LF	68 LF	35.2%
Both measured to outside of siding. The “area” label on the second row appears to be a typo in the county’s own example.			

These feed sheet notes 2 and 3, which cite the Land Use Code Chapter 11 definitions of *remodeling* and *replacement* — a Community Development determination, not a Building Department one. The example does not state the threshold these percentages are measured against, so confirm it with ComDev rather than assuming the City of Aspen’s 40% rule carries over.

ADMINISTRATION

When a depiction plan is required, and what can undo an approval

- **Required for all alterations affecting the existing thermal envelope**, except permits limited to exterior veneer, siding, window or door replacement only.
- **Waivable for minor scopes.** Where the CBO determines the work does not materially impact existing wall assemblies, detailed calculations may be waived under the exception to R103.1.
- **Show existing conditions** and, where applicable, combined existing plus new construction — prepared so the department can clearly verify the total percentage being altered (R103.2.1 requires a depiction of the building thermal envelope).
- **Change orders can retroactively blow the threshold.** A project approved under 75% that later exceeds it by change order must comply with the most recently adopted Pitkin County Energy Code, new-dwelling requirements included. At 68.70%, the example has about 580 sf of envelope in reserve.
- **Siding and window swaps have their own rule.** R503.1.1.1: where all siding and windows are replaced on one or more sides, that envelope must comply with Table R402.1.3, and the applicant must demonstrate the existing cavity insulation R-value to size the continuous insulation.

BEFORE YOU SUBMIT

Drawing checklist

- 01 Three demo calc sheets — roof (flat plane), wall, floor — each carrying the combined running total, not just its own surface.
- 02 Every envelope plane unfolded, lettered, and tagged with gross square footage; letters matching the table exactly.

- 03 Legend with all five conditions: existing to remain, wall to be removed, wall area to be removed, thermal envelope change fill, thermal envelope change line.
 - 04 Asterisked notes explaining any area counted for conversion from non-heated to heated space or for assembly value change.
 - 05 Interior-finish removal areas shown as disturbed envelope, even where framing stays.
 - 06 Windows and doors drawn within panels; square footage on them only where the opening itself is changing.
 - 07 Like-for-like siding walls kept in the denominator with the exclusion note on the sheet.
 - 08 Linear-foot tallies for the Land Use Code determination kept visually separate from the energy calculation.
 - 09 Combined percentage stated plainly, with the roof, wall and floor subtotals shown so the arithmetic can be checked.
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VERIFICATION

Every figure in this document was recomputed from the county's example sheets and reconciles exactly: the 28 wall panels sum to 3,324.01 sf with 2,169.72 sf removed; the three surfaces sum to 9,208.41 sf with 6,326.54 sf removed, giving 68.70%; the starred conversion areas total 981.14 sf; and the linear tallies give 39.9% and 35.2%.

This is a working summary of published county documents, not a code interpretation or a substitute for review by the Pitkin County Building Department. The governing policy memo is dated 1/28/2026 — confirm current adoption and any subsequent amendments before relying on it for a submittal.

SOURCES

Thermal Envelope Example Plan — Pitkin County, sheets A.101–A.103 pitkincounty.com/DocumentCenter/View/35639

Existing Building Thermal Envelope Policy — Jeffrey Erickson, Chief Building Official, 1/28/2026 pitkincounty.com/DocumentCenter/View/35632

Pitkin County Energy Code ordinance — amendments to R502/R503 and REMP pitkincounty.com/DocumentCenter/View/34104

Pitkin County Building Policies pitkincounty.com/206/Building-Policies

Land Use Code Chapter 11 — Definitions pitkincounty.com/DocumentCenter/View/27315